

Understanding the Use of AI-Based Audio Generation Models by End-Users

Jiyeon Han
jy0808@ewhain.net
Ewha Womans University
Seoul, South Korea

Eunseo Yang
y.eunseo@ewhain.net
Ewha Womans University
Seoul, South Korea

Uran Oh*
uran.oh@ewha.ac.kr
Ewha Womans University
Seoul, South Korea

ABSTRACT

With the growing popularity of video platforms, the demand for copyright-free audio sources for adding background music to videos is also expected to increase. While text-to-audio-generation models can be useful for this purpose, little is known about how people perceive and use these models. To understand how generation models for audio are used and to identify their strengths and weaknesses compared to typical audio search engines, we conducted a user study with 16 participants, where they were asked to choose matching background music after watching muted videos. Findings show that participants appreciate the search engine for recommending search keywords and displaying multiple results although the outcome does not fully reflect the intent. In contrast, the generation model posed challenges in choosing proper prompts but excelled in finding the desired music. Based on these results, we suggest design considerations to improve the usability of the audio generation model for end-users.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**; **Usability testing**.

KEYWORDS

empirical study, user experience, text-to-audio, search engine

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1 INTRODUCTION

Considering that the usage of video platforms such as TikTok (tiktok.com) or YouTube (youtube.com) has increased dramatically [3], the demand for copyright-free audio for user-generated content is also expected to grow. However, little has investigated the usability of audio generation models unlike image or text generation models.

Algorithmic composition, also known as automated composition has been researched for a long period [4, 6, 30]. In recent years, artificial intelligence (AI) has been used in generating diverse content types such as images, text, or music, getting significant attention [2, 10, 12]. Variational AutoEncoders (VAEs) [12] or Generative Adversarial Networks (GANs) [9] and Natural Language Processing-based methods such as Long Short-Term Memory (LSTM) [11] or Transformers [27] have developed rapidly and these models have demonstrated the capability to create music that can now compete with music composed by humans. Recently, models like *MusicGen* [5] utilize language models for text-to-music generation, associating text with melodies to generate high-quality music. In addition, text-to-audio models using diffusion models are also being studied, which is *DiffSound* [31]. To enhance the expressiveness of generated music, various approaches are being studied such as applying styles into music [16, 26], adjusting different emotions [20, 22, 33], and controlling audio attributes (e.g., rhythm, instrument types, and note properties) [15, 19, 28]. Inspired by the progress of audio generation models, we aim to enhance the usability of the user interface of audio generation models so that end-users can utilize them easily, and efficiently with high satisfaction.

Reflecting the trend, studies were conducted to understand how these models are perceived by end-users and elevate the usability of these models [17, 21, 25, 29, 32]. Lee and Lin [14], for example, surveyed the usability of image generation models (i.e., *Dall-E2* and *Midjourney*) using the System Usability Scale (SUS). As a result, the interface of *Dall-E2* was clearer and more efficient as it has various features such as the history bar that helps users to search past generated records. Ko et al. suggested design guidelines for text-to-image model. They suggested helping users to input prompts by using prompt engineering and increasing controllability using multi-modal input [13]. As for the text-to-text generation models for texts, Skjuve et al. [24] surveyed early users of *ChatGPT* about their experience, and the results showed that need for prompt writing support is needed as some participants failed to express their requests. In addition, the research investigated how users perceived the differences when using traditional search engines versus *ChatGPT* [23]. The findings indicated that people prefer *ChatGPT* over a search engine because it is perceived as faster, more useful, and user-friendly. While inspiring, little is studied on how to leverage the usability of audio generation models beyond text-to-image or text-to-text models.

We had two research questions: (1) If usage behaviors or the perceived user experience of a search engine and a generation model differ? If so, how? (2) To what extent can existing design guidelines for generation models (e.g., text-to-image) be applied to text-to-audio generation model?

*The corresponding author.

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To identify the challenges and opportunities of audio generation models to find the desired music, we conducted a user study with 16 participants who has little AI knowledge using within-subject mixed method design. They were asked to watch videos without sound and then search for suitable background music using an audio generation model (i.e., *MusicGen*, developed by *Meta*) or an audio search engine (i.e., *Pixabay*, *copyright-free music source website*, one of the popular copyright-free music source websites). In the end, we asked participants about their experiences and collected their feedback including their preferences.

Findings show that participants encountered challenges when entering input prompts for difficulty expressing music in language. To help in creating a prompt, we suggest incorporating keyword suggestions and auto-complete functionalities from search engines into the generation model and introducing platforms for sharing prompt databases similar to text-to-image model. In addition, we recommend showing results from three to five as participants commented that more than five results were overwhelming. Because of the auditory feature of outputs, providing metadata or general characteristics of the audio for the aid of comprehending the audio before listening can help users' decision-making. Finally as in text-to-text generation model, retaining and showing past results could facilitate comparisons and decision-making. The contributions are as follow: (1) the identification of usability issues of existing audio generation models, and (2) improving design suggestions for the user interface of audio generation models for end-users.

2 METHOD

To understand the challenges and the potential use of audio generation models in comparison to audio search engines, we conducted a user study with 16 participants where they were asked to search for background music after watching short muted videos.

2.1 Conditions

As for the comparisons, we used *MusicGen* for the audio generation model and *Pixabay* for the audio search engine. Their screenshot examples are shown in Appendix A. *MusicGen* is a state-of-the-art model developed by *Meta* [5] that generates high-quality music from text descriptions using natural language. We utilized *MusicGen* as the quality of generated music was comparable to the music created by humans with a large language model. *Pixabay* is a website (<https://pixabay.com/music/>) where people upload their images or music as copyright-free. We chose to use this website as a search engine since it has more than 50,000 high-quality music without any additional charges.

2.2 Participants

Sixteen participants (10 female, 6 male) were recruited via an online school community and word-of-mouth from a local community. The inclusion criteria for participation was having a prior experience of adding background music to a video at least once (e.g., video presentation). Their average age was 30.2 ($SD = 2.9$). As for the prior experience with generation models, 11 participants said that they tried text generation models. However, none of them had experience with audio generation models nor background knowledge of AI.

2.3 Apparatus

The study was conducted online using Microsoft Teams. Each participant joined a one-on-one video conference using their computers sharing their screens and audio. The entire session was recorded using a screen recorder with sound. We also used Google Form (forms.google.com) to provide task instructions to the participants, show videos, and collect selected audios for each video. As for the generation model condition, we used *MusicGen*. During the task, participants were not allowed to control the parameters throughout the study and fixed the duration of music to 10 seconds to match the duration of videos. As for the search engine condition, we used *Pixabay*, which has genre or theme recommendations provided by tags. Also, it shows multiple results at a time where each music has titles, album covers created by authors as well as a list of related tags per searched music. As for the videos for the task, we used three different videos: one video for the tutorial and two for the main task. All videos were muted and trimmed to be approximately 10 seconds long. For the tutorial video, we chose a scene from a documentary where a cheetah is hunting prey. The videos used for the main task were selected from commercial advertisements of a watch and a bicycle used for one of the conditions per session in the counterbalanced order. Each videos was selected to ensure participants grasp the atmosphere quickly even when watching short videos without sound.

2.4 Procedure

The study session began by signing the consent form. Then we collected participants' demographic information as well as their experience with adding background music to videos and using AI-based generation models. For the task, participants were asked to watch a 10-second long muted video, generate or search for corresponding music, and make a decision. Participants were instructed to try input prompts multiple times freely to find an appropriate music. For each of the two conditions, a tutorial was given using the same video across the conditions to help participants become familiar with each condition. The presented order of the conditions was counterbalanced and different videos were used for different conditions. At the end of the sessions, we also had an interview to collect participants' experiences and feedback about each condition.

2.5 Data and Analysis

For the analysis, we logged the task completion time of each stage, the number of input attempts, the input word count, and the number of music listened to before making the decision. We also collected 7-scale subjective ratings for easiness, efficiency, and satisfaction for the two conditions as well as their preference. In addition, we asked the reasons for their response as well as the perceived strengths and weaknesses of each condition and coded into themes by two researchers as in [8]. As for the statistical analysis, we used Mann-Whitney U tests to compare the performance between the two conditions since we observed non-normal distribution in our statistical analysis. For the evaluation of 7-scale subjective ratings, the Wilcoxon Signed Rank Test was utilized.

3 FINDINGS

We analyzed the performance and received subjective feedback while performing the task of finding background music for a given video using a generation model and a search engine. The results indicated that the generation model took about twice as long in terms of task completion time and required more attempts compared to the search engine. However, user preferences were evenly divided.

3.1 Task Performance and Usage

3.1.1 Task Completion Time. To compare the task completion time, we measured the time from the moment the search engine or generation model website loaded until the press of the download button of selected music. It averaged 358.0 seconds ($SD=223.1$) when using the generation model, which is significantly longer than the average time of 161.6 seconds ($SD=102.3$) with the search engine; $U = 45, p = .002$. To investigate where most of the time was spent, we examined the task duration per step based on their first attempt (see Figure 1a for the overview of the time).

Input Time. We measured input time to identify how long it takes to input prompts or keywords. It was measured from the loading of the search engine or the generation model website until the press of the search or generate button. It averaged 49.5 seconds ($SD=29.5$) when using the generation model, which is three times longer than the search engine that averaged 15.6 seconds ($SD=11.2$) and this difference was statistically significant; $U = 30, p < .001$.

Response time. Secondly, the response time showed the actual time of generating or showing results. We measured the response time to compare the actual speed to find or generate music. It was measured from the press of the search or generate button to the moment the search engine or generation model showed results. The generation model took 41.7 seconds on average ($SD=5.9$) to show results, and the search engine took 0.2 seconds on average ($SD=0.17$), which is significantly longer; $U = 0, p < .001$.

Decision Time. Lastly, we measured decision time to find out how much it took to decide next action. It was measured from the time when the first result was displayed to the time when either participants pressed the download button after making a decision or started entering a new prompt for the next attempt. When using the generation model, the decision time averaged 20.5 seconds ($SD=6.3$), which is significantly shorter than the average time of 51.5 seconds ($SD=28.4$) when using the search engine; $U = 20, p < .001$.

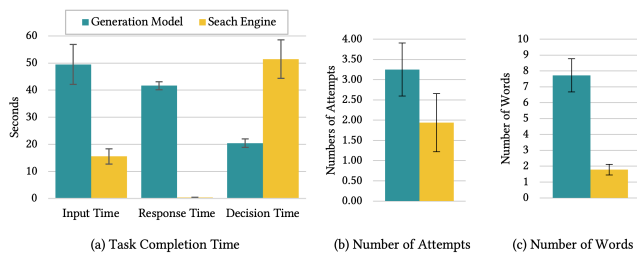


Figure 1: (a) Average task completion time per step for each condition, (b) Average number of attempts, and (c) Average number of words. Error bars indicate standard errors. ($N = 16$)

3.1.2 The Number of Attempts and Reviews. We counted the total number of attempts to input text and the number of music that participants had listened to before making the final decision. By assessing those, we could identify which condition was more efficient to find desired music. The results showed that participants listened to more results for the selection of music although the search engine had fewer input attempts. As shown in Figure 1b the number of attempts averaged 3.25 ($SD=2.14$) for the generation model, which is significantly higher than that of search engine ($M = 1.94, SD=1.29$); $U = 71.5, p = .035$. In addition, the number of music candidates participants listened to before making the final decision was 3.19 on average when using the generation model ($SD=2.17$) and 5.81 ($SD=4.65$) when using the search engine. However, the difference between the two conditions was not significantly different.

3.1.3 The Number of Words for Search Keywords or Prompts. Additionally, we counted the number of words to assess the extent to which participants described music. When comparing the number of words in the input prompts and search keywords, participants entered 7.73 words ($SD=4.18$) when using the generation model, whereas they entered an average of 1.78 words ($SD=1.32$) when using the search engine (please see Figure 1c). Statistical analysis shows that participants used significantly more words when using the generation model than the search engine; $U = 15, p < .001$. For example, P3 entered "good music to listen to while driving in the city" for the generation model, while entered "luxury" when using the search engine. P8 also entered "A sound expressing the combination of small parts" with the generation model but entered "relaxing" with the search engine.

3.2 Perceived User Experience

In this section, we focus on describing the results of the perceived strengths and weaknesses for each condition (see Appendix B for the summary of the results).

3.2.1 Subjective Ratings and Preference. When participants were asked to rate their perceived easiness, efficiency, and satisfaction for the two conditions, both conditions received an average of 4.8 or above on 7-scale ratings, and this difference was not statistically significant according to Wilcoxon Signed Rank Tests. As for the preference, participants' responses were split in half. Seven participants favored the generation model, seven preferred the search engine, and two answered both of them were equally preferable.

3.2.2 Getting Hints for Specific Search Keywords or Input Prompts. On the generation model, many participants appreciated the lack of restrictions on prompt input ($N = 14$). However, it was noted by several participants that it was challenging to come up with appropriate prompts ($N = 11$). In the case of the search engine, as shown in Figure 3a in Appendix B, nine participants mentioned that they liked the ability to modify their search queries based on the tags of the music such as mood and genre, which was displayed in the results. Additionally, P1 mentioned that the search engine provided features like auto-complete and suggested keywords, making it easier to decide suitable keywords.

3.2.3 Descriptiveness of Search Keywords or Input Prompts to Reflect the Intent. Out of the sixteen participants, fourteen mentioned

that they appreciated the flexibility of the generation model which allowed describing the scene of the video in detail and freely inputting all desired elements (e.g., tempo, instrument types). On the other hand, many participants found it challenging to search for music that perfectly matched their desired mood with the search engine ($N = 12$), as indicated by the decision time and the number of music listened to. They mentioned that the search engine required more generalized keywords, as opposed to describing the situation in detail or entering all intents ($N = 12$). Ten participants expressed it was uncomfortable as the search engine sometimes yielded no results when keywords were overly specific. Note that although the search engine used in our study provided search results that included synonyms or similar words for the input keywords, it would not return results if at least one of the keywords entered was not found in the results. In addition, one participant mentioned the inability to finely control musical elements like instruments or tempo as a weakness of the search engine.

3.2.4 The Perceived Time Taken for Getting the Results. As the search engine filters and presents music on the database that matches the keywords, it takes a shorter time to display results. Many participants considered this aspect as a strength of the search engine ($N = 10$). Conversely, several participants pointed out it is a weakness of the generation model in terms of time to display results since the generation model generates music from the input prompt ($N = 9$). However, most of the participants mentioned that generating music was faster than they expected when using the generation model ($N = 10$). One participant stated that the generation model was more accurate and efficient considering the time to compare music from various results, even the generation model is slow to display the result.

3.2.5 The Number of Displayed Results. Regarding the quantity of the results, thirteen participants found the search engine has the advantage that they could compare multiple music in a single search. On the other hand, nine participants expressed they were fatigued occasionally when there were too many results, as they felt compelled to listen to all of the music to find the desired music. Similarly, many participants found it inconvenient that the generation model presented only one result per input, as they could not compare songs ($N = 9$). Nevertheless, four participants appreciated the generation model as they could choose music quickly with one result. Additionally, when asked about the appropriate number of results, participants generally suggested that having between 3 to 5 results would be ideal.

3.2.6 Satisfaction with the Results. For the search engine, eight participants expressed satisfaction with the results, for the fact that humans created the music, which ensured quality. Additionally, seven participants appreciated that the search engine could find similar music even if the keywords were not directly related to them. On the other hand, ten participants answered that the result was satisfactory when all the intents were reflected in the generated music. Among these participants, one expressed satisfaction because he felt a sense of involvement in the process of generating music. However, eight participants expressed dissatisfaction when the model focused more on a specific word of the input, leading to undesired results. One participant suggested the idea of showing

the reflection rate from the generated music. In addition, three participants found it inconvenient when the search engine showed unrelated music, which made it challenging to find the desired one. In the case of the generation model, three participants answered it was dissatisfied with the quality of the music.

3.2.7 Other Opinions. Other than that, participants mentioned strengths of the generation model are the simplicity of the interface ($N = 3$), the feeling of involvement when generating music ($N = 2$), no need for audio editing such as length or speed ($N = 2$), and usable without musical background knowledge ($N = 1$). On the other hand, as weaknesses of the generation model, two participants stated that musical background might be needed to retrieve more desired music. One participant said it was difficult to learn how to use the generation model. Furthermore, two participants mentioned the non-deterministic nature of the generation model. They appreciated the fact that even when entering the same search query, the generation model produced different music. However, one participant among two also pointed out it was sometimes a weakness because the generation model did not store the history of the result which led to the challenge to rediscover music that was previously found. On the other hand, three participants mentioned the strength of the search engine as a familiar interface. In addition, two participants noted that showing the titles of the music was beneficial. One participant considered it as an advantage because they could guess how the music would go without listening to it, while another participant viewed it as a disadvantage for the reason that titles might create bias.

4 DISCUSSION

4.1 Providing Prompt Ideas

The time taken to find the desired music using the generation model compared to the search engine was expected as it takes more to generate music than to fetch the matched existing music. However, it is notable that it took longer to enter the input prompts when using the generation model than the search engine. This may be related to the number of words used in the generation model which was greater than that used in the search engine. According to the participants' feedback, however, it could be mainly due to the difficulty in coming up with prompt ideas because expressing music with technical terminology that leads to expressing accurate nuances is known to be difficult when users have limited musical knowledge [7]. Considering that participants appreciated the keyword suggestion and auto-complete features of the search engine, introducing such features into the generation model could be a way to enhance task efficiency. In addition, if the short example audio of each suggested keyword is provided, it could shorten the time to come up with input prompts. Furthermore, the text-to-audio generation model could benefit from having databases or communities for sharing prompts (e.g., PromptPal (promptpal.net)) as in text-to-image generation models.

4.2 Setting Keyword Reflection Ratios

As highlighted in subsection 3.2.6, participants' satisfaction with the results was largely affected by how well the generation model reflected each word they used to describe the desired music in the

prompt. Moreover, the flexibility of what can be entered in the prompt was the perceived strength of the generation model mentioned by the most number of participants. Yet, some participants felt that some of the words had more weight than others when listening to the output although they wished every word to be evenly reflected to the results. One possible solution is to show the reflection ratio of the prompt input on the generated result as suggested by Arrieta et al. [1], which could assist users in modifying the prompt. In addition, allowing users to directly specify the reflection rate of each input word similar to setting a parameter could be another idea to guarantee satisfying results.

4.3 Displaying Multiple Results

While most participants said that satisfied with the results of the generation model with more descriptive input, findings also show that participants made more modifications to the generation model than the search engine. One possible reason for this result is the number of displayed results. That is, while participants can listen to multiple music with a single query and choose the best one they like after making comparisons among the search results using the search engine, the participant has to try re-prompting again if they are not satisfied with the single result generated by the generation model. Meanwhile, participants did not like having too many results for the reason that it was hard to listen to all of the displayed ones for making decisions. Considering the prior findings generating results with 3-9 different seeds is ideal for image generation models by Liu and Chilton. [18] and the ideal number of results that participants' responses were 3 to 5, we suggest showing multiple results but with a limited number while improving the accuracy to reduce the burden of listening to all the generated music for comparisons.

4.4 Describing the Generated Audio Files

While some participants showed concern that titles or album covers could potentially lead to biases, others valued having titles, album cover images, and tags for the results using the search engine was helpful for them to understand what to expect before listening to each music since there is no immediate way to grasp what audio entails, unlike images or text. Therefore, metadata or hashtags that describe the audio could be useful for the generation model as well. One approach to describe audio objectively without potential bias is to utilize an audio-to-text model to analyze the metadata of the audio, such as its overall characteristics, tempo, and types of instruments used in the music.

4.5 Keeping Track of Generated Output

Our findings show that participants tend to make several attempts and listen to more than one piece of music to make a decision. Moreover, the number of attempts was significantly higher with the generation model than the search engine. Yet, unless a local copy was made, it was not possible to access the previous results with the generation model even with the identical prompt as the model was not deterministic. Even for the deterministic model, it takes a long time to regenerate the desired music that is once generated. Thus, we recommend keeping the records of the generated output so that users can review and make comparisons as the history bar of past records suggested by Lee and Lin [14]. In addition, we suggest

providing a memo box in the record to review the features of past results without listening.

4.6 Limitations and Future Work

In our user study, we utilized *Pixabay*, one of the most commonly used royalty-free music websites, as the search engine. However, *Pixabay* may not represent all audio search engines, and some feedback may be specific to this particular search engine. Similarly, the generation model used in the user study, *MusicGen*, may not include all features of audio generation models, and the interface may vary depending on the model. Thus, some of the feedback related to the generation model may also be limited to this specific model. In addition, our recruitment focused on participants with prior experience in adding audio to videos. However, depending on the users who have no related experience or users with specialized musical knowledge, the results could be different. In addition, despite the enough time to be familiar with each condition, participants were generally used to search engines that can affect the results. In the future, we plan to conduct research aimed at designing an interface for text-to-audio generation models, reflecting the proposed methods based on the findings, to enable users to generate the desired audio more efficiently.

5 CONCLUSION

The popularity of video platforms such as YouTube is on the rise, leading to an increased demand for audio content that does not have copyright restrictions. Despite the advancements, research about the usability of the audio generation models is little, unlike image or text generation models. In this context, we conducted a user study to investigate how users search audio using the audio generation model in comparison to audio search engines. The results showed that although it took more time to find background music using generation models, users had the same preferences for both generation models and search engines. Participants praised the search engine's ability they can compare music quickly, but also they mentioned it is tiring to listen and compare all the results. In contrast, participants could describe the mood or situation of the music on the prompt despite the challenge of coming up with input keywords and occasional dissatisfaction with the results. The paper contributes by identifying usability issues in audio generation models and suggesting design improvements for end-users.

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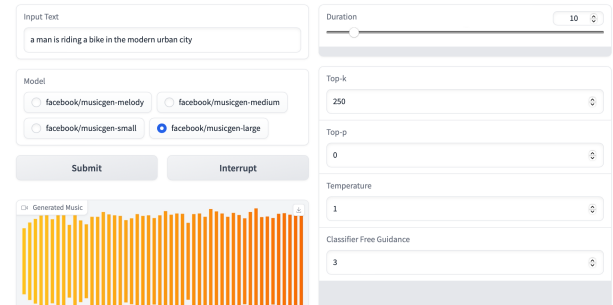
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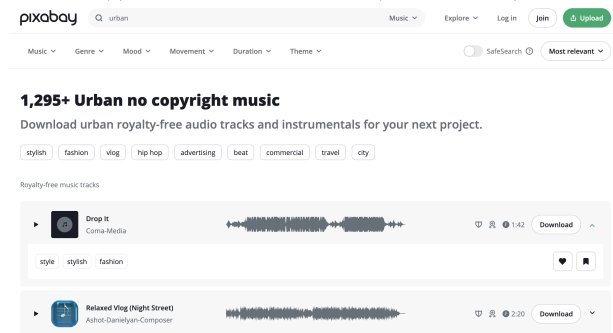
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APPENDIX

A SCREENSHOT EXAMPLE OF MUSICGEN AND PIXABAY



(a) Audio Generation Model (i.e., MusicGen)



(b) Audio Search Engine (i.e., Pixabay)

Figure 1: Screenshot examples of (a) MusicGen and (b) Pixabay showing the generation or search results. While MusicGen shows one result at a time, Pixabay displays multiple results.

B SUMMARY OF THE PERCEIVED USER EXPERIENCE

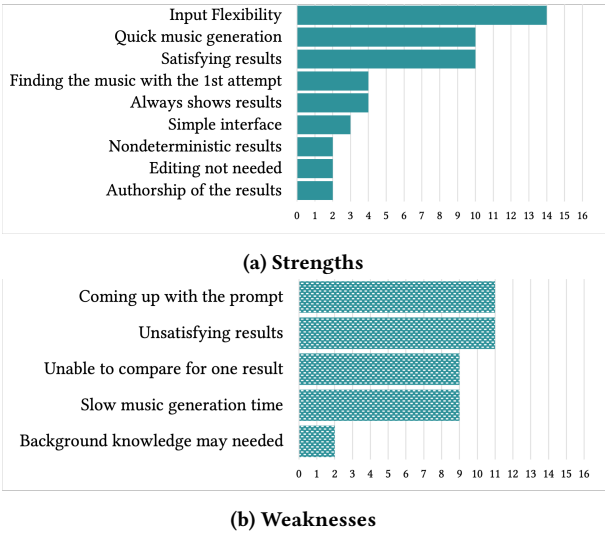


Figure 2: Perceived (a) strengths and (b) weaknesses of the generation model mentioned by more than one participant, sorted by the number of responses ($N = 16$).

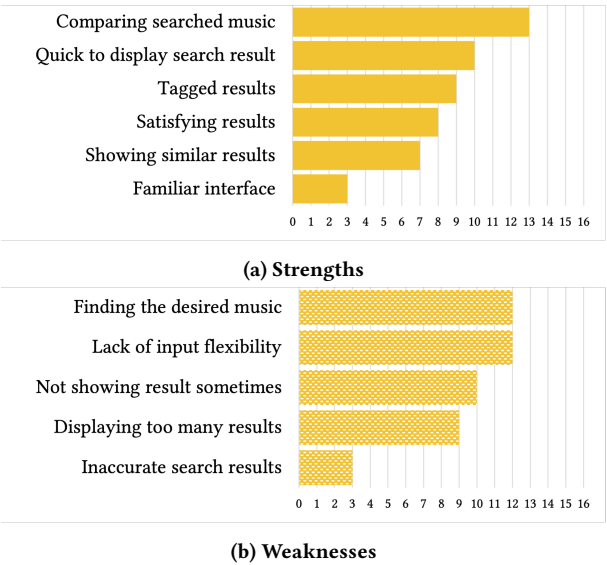


Figure 3: Perceived (a) strengths and (b) weaknesses of the search engine mentioned by more than one participant, sorted by the number of responses ($N = 16$).